

Composite_Model_Documentation

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Change History

Version	Changes
1.1	Added materialvariantlocked as a new property for model type entities.

1 Introduction

Composite model files exist as *.composite files in the art directory. The purpose of these files is to combine models, decals, effects, lights, and extra attachment points into a single “composite” model. These model types can be referenced in AnimXML files. In some cases, composite models may be spawned in directly, e.g., via a destruction model that completely shatters.

These files are stored in a JSON file format.

Composite Models are a template for Composite Instances in the game.

2 Composite File Definition

2.1 Top-Level Properties Overview

Composite models consist of the following top-level properties that contain sub-properties or arrays of additional data. All objects are optional. The composite definition will fall safely back to defaults if they are excluded.

JSON Object Name	Description
entities	An array of all entities owned by the composite model, along with properties that define additional properties that may modify how the entities appear.
events	An array of events that may trigger actions during the lifetime of a model.
destructionsharedsettings	Destruction-related additional settings.
globalsettings	Stores a list of sub-properties as settings that may affect all entities for the composite model.

2.2 Entities

2.2.1 Shared Entity Properties

These properties are read in for all entity types. However, they do not necessarily have a significant impact on behavior in all cases.

JSON Object Name	Description
local_transform	3 rows x 4 columns matrix that is crushed into a list of comma-separated values. If this is not specified, the matrix will default to identity. The local_transform represents the offset for the entity root in local space relative to the composite model's root.
probability	An optional value that gives a chance for the object to appear in an instance. Defaults to 1.0 if not defined. Values between 0 and 1 are expected, with the value representing the chance.

	If the uniformly generated random number between 0 and 1 is greater than the probability, it will cull the entity during instance generation. NOTE: This functionality is not currently enabled in-game.
seed	Defines the int to use as the seed given to the random generation probability. NOTE: This functionality is not currently enabled in-game.
passive_rotation	Animates with speed on any axis with passive rotation greater than 0. Defined as a list of “x, y, z” passive rotation speeds.
enable_lod	Boolean. Turns on the EnableLOD feature.
maintain_terrain_height	Boolean. Turns on the MaintainTerrainHeight feature.
maintain_terrain_orientation	Boolean. Turns on the MaintainTerrainOrientation feature.
is_ghost	Boolean. Turns on the IsGhost status. This means that the visual will not be rendered. Typically, these are used as helpers to block out more complex placements.
deathbehaviour	When not defined: * Death behavior is set to DefaultAutoHideUnlessDestructionTemplate * Death percent is set to 0.0f. Expects a string defining the type of death logic to use: • "Default" • "KeepAfterDeath" • "HideOnDeath" • "PhysicsOnDeath" • "HideAt75" • "HideAt55" • "HideAt35" • "HideAt15" • "HideAtCustom" • "PhysicsAt75" • "PhysicsAt55"

- "PhysicsAt35"
- "PhysicsAt15"
- "PhysicsAtCustom"

Information about values:

- Default forces a hide of the entity only when the object is considered dead. Unless a destruction model is defined, it will trigger physics upon death.
- KeepAfterDeath makes the entity continue showing after death.
- HideOnDeath causes the entity to hide when the unit hits 0% health immediately.
- PhysicsOnDeath causes the entity to activate physics for the object on death.
- HideAt75, HideAt55, HideAt35, HideAt15 will cause the entity to hide at the specified health percent and lower.
- HideAtCustom uses the value defined by an additional deathbehaviour_custom to determine when to hide.
- PhysicsAt75, PhysicsAt55, PhysicsAt35, PhysicsAt15 will cause the entity to activate physics at the specified health percent and lower.
- PhysicsAtCustom uses the value defined by an additional deathbehaviour_custom to determine when to activate physics.

deathbehaviour_custom

Only does something if deathbehaviour is defined. Expected if you set the deathbehavior property to "HideAtCustom" or "PhysicsAtCustom". When either of these is set and this additional property is not defined, the entity will default this value to 0.0f.

Expects a float between 0 and 100 representing the percent of health to trigger at.

destruction_customdestruction

Expects a boolean. No behavior in-game. Used for hiding and showing the template options on an entity. This setting is only relevant for development tools.

destruction_dontusetemplate	Expects a boolean. Forces an override to prevent normal destruction instance creation in-game. E.g., if you don't want something to swap to the destruction version of the TMMModel.
destruction_manualltargetchoice	Expects a boolean. Deprecated property from an older version of destruction.
destruction_ignore	Expects a boolean.
destruction_disabletargetlow	Expects a boolean. Deprecated property from an older version of destruction.
destruction_disabletargettitan	Expects a boolean. Deprecated property from an older version of destruction.
destruction_disabletargethigh	Expects a boolean. Deprecated property from an older version of destruction.
destruction_dontusephysicstemplate	Expects a boolean. Prevents the object from using the simple physics object that may be defined. E.g., Barrels have a simple physics object defined in their TMMModel, which is different from the model having destruction chunks.
destruction_destroymode	Expects one of the following strings when defined: • "default" • "anytimedestroy" • "onfulldestroyonly"
destruction_hitstodestroy	Defaults to 1 when not specified. Expects a greater than zero integer that defines the number of times the object must be hit before it is destroyed.

2.2.2 Model Type Entity

Models refer to TMMModel assets. The properties below are unique to the Model type entity in addition to those listed in 2.2.1. Note that only Model type objects can use destruction templates, where those are referred to in other places.

JSON Object Name	Description
model_ref	Expects a valid relative path string to a TMMModel. Expected value, and you will not see anything if this is excluded or the

	path is invalid.
anim_ref	Expects a valid relative path string to a TMAAnimation. (optional - Defaults to none if not set or if the path is invalid). Plays the specified animation on the model.
materialvariant	Expects an integer defining the material variant index to use. Defaults to 0 if not specified. Does nothing in-game unless the materialvariantlocked value is set to true.
materialvariantlocked	Expects a boolean. When set to true, it forces the use of the “materialvariant” set above. This can be useful if you want a specific variation included in a model.
submodel	Expects a string form name specifying the submodel to force. This allows you to show chopped trees, for example. Will only do something if the submodel is defined inside the TMMModel. Behind the scenes, this looks up the index by name and will default to 0 and an empty string when not specified.

2.2.3 Decal Type Entity

Decal-type assets are used to apply textures as part of the model. The properties below are unique to the Decal type entity in addition to those listed in 2.2.1. Decals work as a projected texture that projects in 3d space.

JSON Object Name	Description
edge_fade	Expects a float. Fades for distance around the edge of the texture.
bounds_fade	Expects a float. Fades for distance around the bounds.
normal_fade	Expects a float. Fades for distance applied to normals.
alpha_cutout	Expects a float. Modifies the alpha cut-out property.
sort_index	Expects an int. Controls the layering order of textures to force a priority of which will appear on top.
texture_scale_x	Expects a float. Rescales the X axis of the texture.
texture_scale_y	Expects a float. Rescales the Y axis of the texture.

texture_offset_x	Expects a float. Offsets the texture on the X axis.
texture_offset_y	Expects a float. Offsets the texture on the Y axis.
texture_animation_offset_x	Expects a float. Defines texture animation movement on the X axis.
texture_animation_offset_y	Expects a float. Defines texture animation movement on the Y axis.
basecolor_tex	String reference to a texture file.
normal_tex	String reference to a texture file.
masks_tex	String reference to a texture file.
emissive_tex	String reference to a texture file.
unscaled_alpha_mask	String reference to a texture file.
emissive_intensity	Expects a valid float to control the intensity of the emissive. 0.0f to 10.0f is a recommended range.
emissive_color	Expects an array of floats with “r, g, b” as a color. Stored as values between 0 and 1 for each color.
mask_mode	Expects a string with the following valid settings: • "default" • "terrain_only" • "buildings_and_terrain" • "buildings_only" Note: although it says “Buildings,” this really means ANY model. Default will apply the projected texture onto everything.
useglobalshadercolor	Boolean. When set to true, this allows the texture to modify based on any color that is applied to a whole visual. When set to false (default), it will ignore those colors.
useglobalshaderalpha	Boolean. When set to true, this allows the texture to multiply its alpha by the value defined for the whole model. (e.g., when the model is progressively fading out on death.) Defaults to false.

2.2.4 Popcorn Type Entity

Popcorn refers to PopcornFX assets. The properties below are unique to the Popcorn type entity in addition to those listed in 2.2.1. Popcorn assets are VFX such as puffs of smoke or fire.

JSON Object Name	Description
pkfx	Expects a string defining the relative path to the asset.

2.2.5 Point Light Type Entity

Point Lights are simple lights that you can define within the composite. The properties below are unique to the Point Light type entity in addition to those listed in 2.2.1.

JSON Object Name	Description
color	Expects an array of floats with “r, g, b” as a color.
intensity	Expects a float representing the intensity of the light.
range	Expects a float representing the range of the light.

2.2.6 Attach Point Type Entity

Attach Point entities have no visual appearance. They instead act as a way to create a fake “bone” for referencing as an attachment point inside the local space of a composite model. The properties below are unique to the Attach Point type entity in addition to those listed in 2.2.1. The way these work is very similar to how they work for TMMModels.

JSON Object Name	Description
attachname	String. Name of the attach point.
preview_model_ref	String. Relative path to a model or empty. Acts as a ghost model in the in-house model viewer tool. Does not show in-game.
DummyBoneMode	Optional. Enables forced Dummy Bone Mode to control what happens with the Attachment Name. Valid values: • "Auto" • "DoNotDummyBone" • "AddToSim"

	• "ForceToDummyBoneType" Information about modes: Controls whether the bone is added to the simulation. Auto - AddToSim if detected to be a dummy bone type (otherwise ignored). DoNotDummyBone - Ignores for Sim. AddToSim - Adds to the Sim (even if not paired as a dummy bone). ForceToDummyBoneType - Forces to a specified dummy bone type.
ForcedDummyBoneName	When setting the DummyBoneMode to ForceToDummyBoneType you should also specify this string property to set the name to treat the bone as. E.g., to treat it as a Relic attach point. This is similar to what can be done for attachment points inside TMMModels.

2.3 Events

Composite events provide a very primitive way of applying a limited number of actions for a limited number of simple situations when events are allowed to trigger. These are intended as a cheap way to provide ways to control how visuals fade in/out as part of a sequence of events.

Example:

```

1  "events": [{
2      "trigger": "OnSpawn",
3      "action": "FadeOutAllDecals",
4      "duration": 0.0,
5      "nextcustom": "Wait2Start"
6  }, {
7      "trigger": "OnCustom",
8      "triggername": "Start",
9      "action": "FadeInAllDecals",
10     "duration": 0.5,
11     "nextcustom": "Hold"
12  }, {
13     "trigger": "OnCustom",
14     "triggername": "Hold",
15     "action": "WaitForTime",
16     "duration": 1.0,
17     "nextcustom": "End"
18  }, {
19     "trigger": "OnCustom",
20     "triggername": "End",
21     "action": "FadeOutAllDecals",
22     "duration": 0.20000000298023225
23  }, {

```

```

24     "trigger": "OnCustom",
25     "triggername": "Wait2Start",
26     "action": "WaitForTime",
27     "duration": 0.30000001192092898,
28     "nextcustom": "Start"
29 },

```

2.3.1 Event Properties

Property	Type	Description
trigger	String	See 2.3.3 for a list of valid values and their purposes.
triggername	String	Name to allow calling a trigger using the “nextcustom” property.
action	String	Action to perform during the duration, starting at 0s and ending when the time is up. See 2.3.2 for a list of actions and what they do.
actionproperty	String	Additional value to use as part of an action. Currently, it only applies to the PlaySoundSet action and specifies the Sound Set name to play
duration	Float	The time in seconds for the action to play over. At the end of the duration the nextcustom property will be triggered if it is set.
nextcustom	String	Optionally defines a string for the next event to call in sequence. Note that you can use non-unique trigger names to call multiple simultaneous actions.

2.3.2 Event Action Types

Type	Description
None	Performs no action.
FadeInAllDecals	Fades Decals from 0% to 100% alpha as a multiplier during duration.
FadeInAllModels	Fades Models from 0% to 100% alpha as a multiplier during duration.
FadeOutAllDecals	Fades Decals from 100% to 0% alpha as a multiplier during duration.
FadeOutAllModels	Fades Models from 100% to 0% alpha as a multiplier during duration.
WaitForTime	Waits for the set duration, doing nothing.

PlaySoundSet	Plays a specified sound set defined in “actionproperty” and then waits for the duration.
FadeAllTreesToBurnt	Fades all child tree objects from 0% burnt to 100% burnt during the duration.

2.3.3 Event Trigger Types

Type	Description
None	Acts as a disabled trigger type.
OnSpawn	Triggers when the unit first spawns.
OnShatter	Triggers when the unit dies.
OnCustom	Relies on other actions to trigger them using the nextcustom property.

2.4 Destruction Shared Settings

Destruction settings are a special subset of settings related to how global properties for destruction are handled.

Example:

```
1 "destructionsharedsettings": {
2   "ignorefordestruction": false
3 },
```

Property	Type	Description
ignorefordestruction	boolean	Used to disable destruction event propagation to all entities owned by the composite model. You can apply an individual entity version of this

2.5 Global Settings

The following are defined in a global settings object as additional properties that can change the behavior of composite models.

Property	Type	Description
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useboundsduringrectselection	Boolean	When enabled, this will check the bounding box for collision during a screen rect selection and consider it a hit if the box is hit. This should be used sparingly and is typically reserved for composite models that contain no other models.
selectionpadding	Float Array	Pads the tight bounds to make it easier to select the object. Defined as an array of X, Y, Z bounds.
trueboundspadding	Float Array	Pads the visual bounds to make it more likely the renderer will consider the model valid. (E.g. popcorn effects don't have bounds so may need this inflated). Defined as an array of X, Y, Z bounds.
useglobalalphaonalldecals	Boolean	When enabled, it forces alpha to be set on all decals each render. Can be turned on for individual entities if desired.
useglobalcoloronalldecals	Boolean	When enabled, it forces color to be set on all decals each render. Can be turned on for individual entities if desired.
animatemodels	String Array	List of models to forward animations to.
globalhideduringcinematicmode	Boolean	When enabled, it hides the model when it enters named modes. In this case, it will hide when a Cinematic is playing.
globalhideduringpregamemode	Boolean	When enabled, it hides the model when it enters named modes. In this case, it will hide in the main menu.
globalhideduringnouimode	Boolean	When enabled, it hides the model when it enters named modes. In this case, it will hide when the UI is toggled off.